

Quiz 2

Student Name: _____

Student ID: _____

A continuous random variable Y has the probability density function

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Tasks (5 points each)

1. Verify that $f(y)$ is a valid probability density function.

2. Compute the mean $E[Y]$.

3. Determine the probability that Y is less than its mean.

4. Sketch (roughly) the shape of the PDF and the CDF(cumulative) **on the same axes**, labeling all key points.